



ADVANCED RESEARCH METHODOLOGY Weekly Lecture Schedule • Fall 2026

MS Information Technology • PhD Information Technology • Evidence, Design and Integrity in Computing Research

CREDIT HOURS		DURATION	TRACKS
3 (3–0): no laboratory hour		16 teaching weeks; mid-term Week 8	MS IT and PhD IT
📅	📖 Chapters and topics	👤 In class	📁 Portfolio due
1	1 What Counts as a Contribution – kinds of contribution, novelty 2 Philosophy of Science – falsification, five research paradigms	What a contribution is; the paradigms behind method choice	—
2	3 Questions, Hypotheses, Constructs – FINER, PICOC, alignment 4 The Publication System – HJRS categories, peer review	FINER and PICOC on the students' own topics; HJRS categories	P1 Topic statement
3	5 Systematic Literature Review and Mapping – the protocol, search, screening, appraisal, extraction and synthesis, the PRISMA flow	Protocol workshop: search strings built and run live	—
4	6 Snowballing, Grey Literature – scoping reviews, publication bias 7 Bibliometrics and References – citation analysis, reference tools	Snowballing; Zotero and VOSviewer at the keyboard	P2 SLR protocol Quiz 1
5	8 Reading and Appraising a Paper – three-pass reading, refereeing 9 Design Science Research – artefact types, Hevner, DSRM, FEDS	Three-pass reading of one assigned paper; DSR artefact types	—
6	10 Controlled Experiments – three designs, pilots, human subjects 11 Quasi-Experiments and Mining – selection bias, causation	Designing a code-review experiment; confounding in repository data	—
7	12 Case Study Research – propositions, case selection, claims 13 Survey Research – sampling, nonresponse, writing items	Unit of analysis; instrument critique	P3 Design choice memo
8	Mid-term examination – Chapters 1–13, 30 marks, 1 h 15 14 Qualitative Methods and Coding – interviews, saturation, coding	Mid-term examination, then coding practice	Mid-term
9	15 Mixed Methods – four designs, integration, triangulation 16 Benchmarking and Simulation – suites, non-determinism	Joint displays; workload characterisation	—
10	17 Measurement and Validity – scales, reliability, validity 18 Sampling and Study Power – effect sizes, underpowered studies	Operationalising a construct; a power calculation each	P4 Draft instrument
11	19 Statistical Inference Done Properly – what a p-value is, estimates not verdicts, multiple comparisons, assumptions	The p-value misinterpretation clinic; effect sizes in R	Quiz 2
12	20 Regression for Inference – coefficients, collinearity, mixed effects 21 Structural Equation Modelling – CB- or PLS-SEM, method bias	Model interpretation; a PLS-SEM run in <i>semnr</i>	—
13	22 Evaluating ML Research – eight ways information leaks, baselines 23 Qualitative Data Analysis – from code to claim, audit trail	A leakage audit on a supplied notebook; coding reliability	P5 Analysis plan
14	24 Threats to Validity – four questions, threats by type, writing it 25 Reproducibility and Artefacts – deposit, FAIR data, licensing	Writing a threats section; packaging an artefact	Quiz 3
15	26 Pre-registration – registered reports, questionable practices 27 Research Ethics – consent, anonymisation, dual use 28 Academic Integrity – the rules, paraphrasing, authorship 29 Generative AI in Research – disclosure, fabricated citations	The rigour and integrity block; similarity report interpretation	P6 Pre-registration
16	30 The Synopsis and Proposal – structure, the deciding sections 31 Writing the Thesis and Paper – IMRaD, introduction funnel 32 Publishing and Peer Review – venue, cover letter, reviewers 33 Presenting and Defending – three examinations, the defence talk	Production workshop and defence-style presentations	P7 Synopsis draft Presentation

ASSESSMENT WEIGHTS, FIXED BY REGULATION

5%	5%	5%	5%	30%	50%
Participation	Quizzes: Wk 4, 11, 14	Assignment: P1–P7	Presentation, Week 16	Mid-term, Week 8	Final-term, all chapters

I Foundations of Inquiry in Computing
IV Measurement, Data and Analysis

II Finding and Synthesising Knowledge
V Rigour, Reproducibility and Ethics

III Research Designs for Computing
VI Producing and Defending Research

Chapter numbers and full titles are in the Fall 2026 Advanced Research Methodology handout (Appendix H, Sixteen-Week Schedule and Assessment Map). P1–P7 are one continuous piece of work, each a section of the synopsis you will submit: MS IT scholars target a thesis synopsis, PhD IT scholars a dissertation synopsis with a first publishable increment identified. Weights follow PhD Regulations 2024 §51 and MPhil/MS Regulations 2024 §36.